Inference for numerical data

Daanish Rasheed

Getting Started

Load packages

In this lab, we will explore and visualize the data using the **tidyverse** suite of packages, and perform statistical inference using **infer**. The data can be found in the companion package for OpenIntro resources, **openintro**.

Let's load the packages.

```
library(tidyverse)
library(openintro)
library(infer)
```

The data

Every two years, the Centers for Disease Control and Prevention conduct the Youth Risk Behavior Surveillance System (YRBSS) survey, where it takes data from high schoolers (9th through 12th grade), to analyze health patterns. You will work with a selected group of variables from a random sample of observations during one of the years the YRBSS was conducted.

Load the `yrbss` data set into your workspace.

```
data('yrbss', package='openintro')
```

There are observations on 13 different variables, some categorical and some numerical. The meaning of each variable can be found by bringing up the help file:

```
?yrbss
```

1. What are the cases in this data set? How many cases are there in our sample?

Insert your answer here

Each case is a single high school student who responded to the YRBSS survey. There are 13,583 cases in the sample, each with 13 variables.

Remember that you can answer this question by viewing the data in the data viewer or by using the following command:

```
glimpse(yrbss)
```

```
## Rows: 13,583
## Columns: 13
## $ age                   <int> 14, 14, 15, 15, 15, 15, 15, 14, 15, 15, 15, 1~
## $ gender                <chr> "female", "female", "female", "female", "fema~
## $ grade                 <chr> "9", "9", "9", "9", "9", "9", "9", "9", "9", ~
## $ hispanic              <chr> "not", "not", "hispanic", "not", "not", "not"~
## $ race                  <chr> "Black or African American", "Black or Africa~
## $ height                <dbl> NA, NA, 1.73, 1.60, 1.50, 1.57, 1.65, 1.88, 1~
```

```
## $ weight                   <dbl> NA, NA, 84.37, 55.79, 46.72, 67.13, 131.54, 7~
## $ helmet_12m               <chr> "never", "never", "never", "never", "did not ~
## $ text_while_driving_30d   <chr> "0", NA, "30", "0", "did not drive", "did not~
## $ physically_active_7d     <int> 4, 2, 7, 0, 2, 1, 4, 4, 5, 0, 0, 0, 4, 7, 7, ~
## $ hours_tv_per_school_day  <chr> "5+", "5+", "5+", "2", "3", "5+", "5+", "5+",~
## $ strength_training_7d     <int> 0, 0, 0, 0, 1, 0, 2, 0, 3, 0, 3, 0, 0, 7, 7, ~
## $ school_night_hours_sleep <chr> "8", "6", "<5", "6", "9", "8", "9", "6", "<5"~
```

Exploratory data analysis

You will first start with analyzing the weight of the participants in kilograms: `weight`.

Using visualization and summary statistics, describe the distribution of weights. The `summary` function can be useful.

```
summary(yrbss$weight)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NAs
##   29.94   56.25   64.41   67.91   76.20  180.99    1004
```

2. How many observations are we missing weights from?

Insert your answer here

```
sum(is.na(yrbss$weight))
```

```
## [1] 1004
```

We are missing weights for 1,004 observations.

Next, consider the possible relationship between a high schooler's weight and their physical activity. Plotting the data is a useful first step because it helps us quickly visualize trends, identify strong associations, and develop research questions.

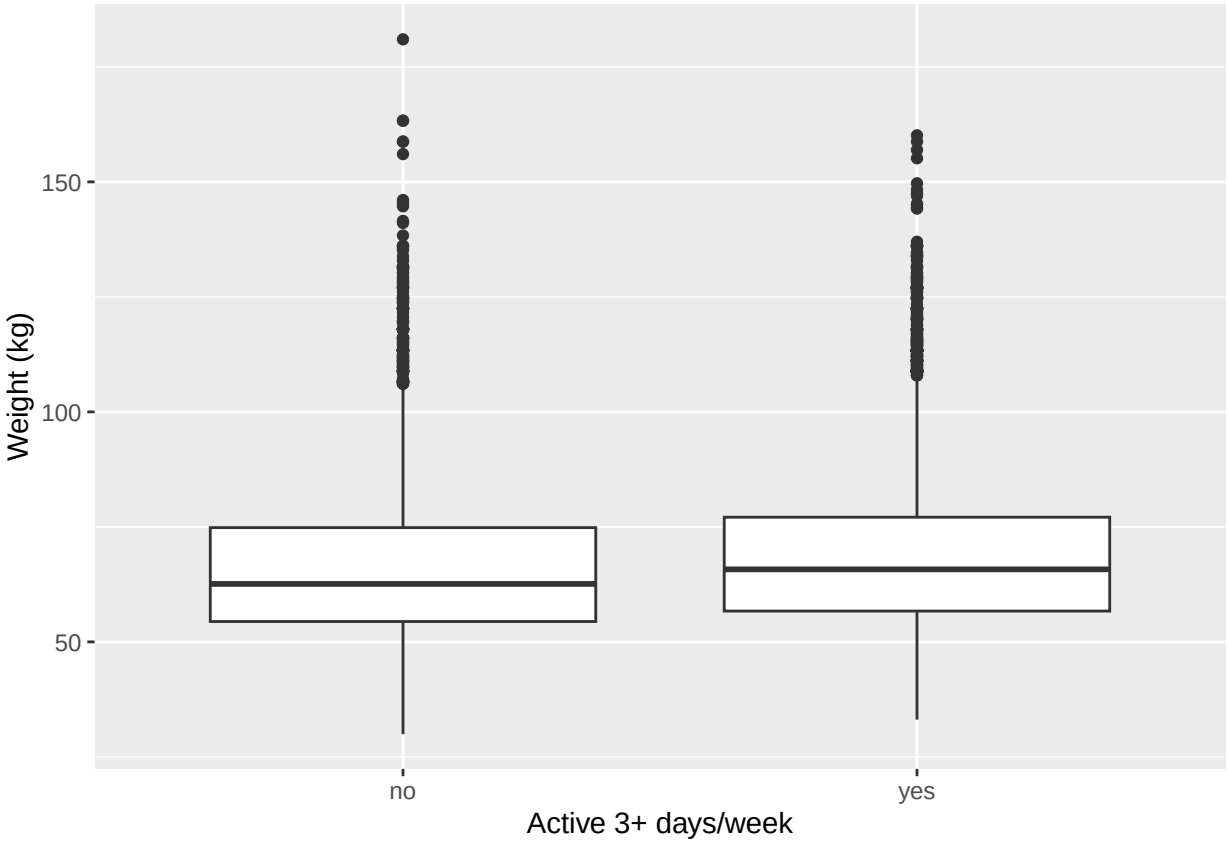
First, let's create a new variable `physical_3plus`, which will be coded as either "yes" if they are physically active for at least 3 days a week, and "no" if not.

```
yrbss <- yrbss %>%
  mutate(physical_3plus = ifelse(yrbss$physically_active_7d > 2, "yes", "no"))
```

3. Make a side-by-side boxplot of `physical_3plus` and `weight`. Is there a relationship between these two variables? What did you expect and why?

Insert your answer here

```
yrbss %>%
  drop_na(physical_3plus) %>%
  ggplot(aes(x = physical_3plus, y = weight)) +
  geom_boxplot() +
  labs(x = "Active 3+ days/week", y = "Weight (kg)")
```



Both box plots are quite similar. The median for the "yes" category is a little bit higher, but both plots are positively skewed, showing a lot of high outliers. So, at best, there is a slight association between the two variables. This relationship should have gone the other way since my prediction was that students who do exercise are lighter. The reason behind this is that muscle is heavier than fat, and students who do sports tend to be larger in size.

The box plots show how the medians of the two distributions compare, but we can also compare the means of the distributions using the following to first group the data by the `physical_3plus` variable, and then calculate the mean `weight` in these groups using the `mean` function while ignoring missing values by setting the `na.rm` argument to `TRUE`.

```
yrbss %>%
  group_by(physical_3plus) %>%
  summarise(mean_weight = mean(weight, na.rm = TRUE))
```

```
## # A tibble: 3 x 2
##   physical_3plus mean_weight
##   <chr>                <dbl>
## 1 no                    66.7
## 2 yes                   68.4
## 3 <NA>                  69.9
```

There is an observed difference, but is this difference statistically significant? In order to answer this question we will conduct a hypothesis test.

Inference

4. Are all conditions necessary for inference satisfied? Comment on each. You can compute the group sizes with the `summarize` command above by defining a new variable with the definition `n()`.

Insert your answer here

```
yrbss %>%
  drop_na(physical_3plus, weight) %>%
  group_by(physical_3plus) %>%
  summarise(mean_weight = mean(weight), sd_weight = sd(weight), n = n())
```

```
## # A tibble: 2 x 4
##   physical_3plus mean_weight sd_weight     n
##   <chr>                <dbl>     <dbl> <int>
## 1 no                    66.7      17.6  4022
## 2 yes                   68.4      16.5  8342
```

Independence**:** The YRBSS is a random sample of high schoolers, and one student's weight shouldn't affect another's. The two groups are separate students, so they are independent of each other. The sample is far smaller than 10% of all U.S. high schoolers.

Sample size / skew: The groups are large (about 8,342 "yes" and 4,022 "no" with recorded weights). The weight distributions are right-skewed, but with samples this big the Central Limit Theorem makes the sampling distribution of the difference in means approximately normal.

5. Write the hypotheses for testing if the average weights are different for those who exercise at least times a week and those who don't.

Insert your answer here

Let μ_{yes} be the mean weight of students who are physically active at least 3 days per week and μ_{no} the mean weight of those who are not.

- $H_0 : \mu_{yes} - \mu_{no} = 0$ - there is no difference in average weight between the two groups
- $H_A : \mu_{yes} - \mu_{no} \neq 0$ - the average weights differ

Next, we will introduce a new function, `hypothesize`, that falls into the `infer` workflow. You will use this method for conducting hypothesis tests.

But first, we need to initialize the test, which we will save as `obs_diff`.

```
obs_diff <- yrbss %>%
  drop_na(physical_3plus) %>%
  specify(weight ~ physical_3plus) %>%
  calculate(stat = "diff in means", order = c("yes", "no"))
```

Notice how you can use the functions `specify` and `calculate` again like you did for calculating confidence intervals. Here, though, the statistic you are searching for is the difference in means, with the order being `yes - no != 0`.

After you have initialized the test, you need to simulate the test on the null distribution, which we will save as `null`.

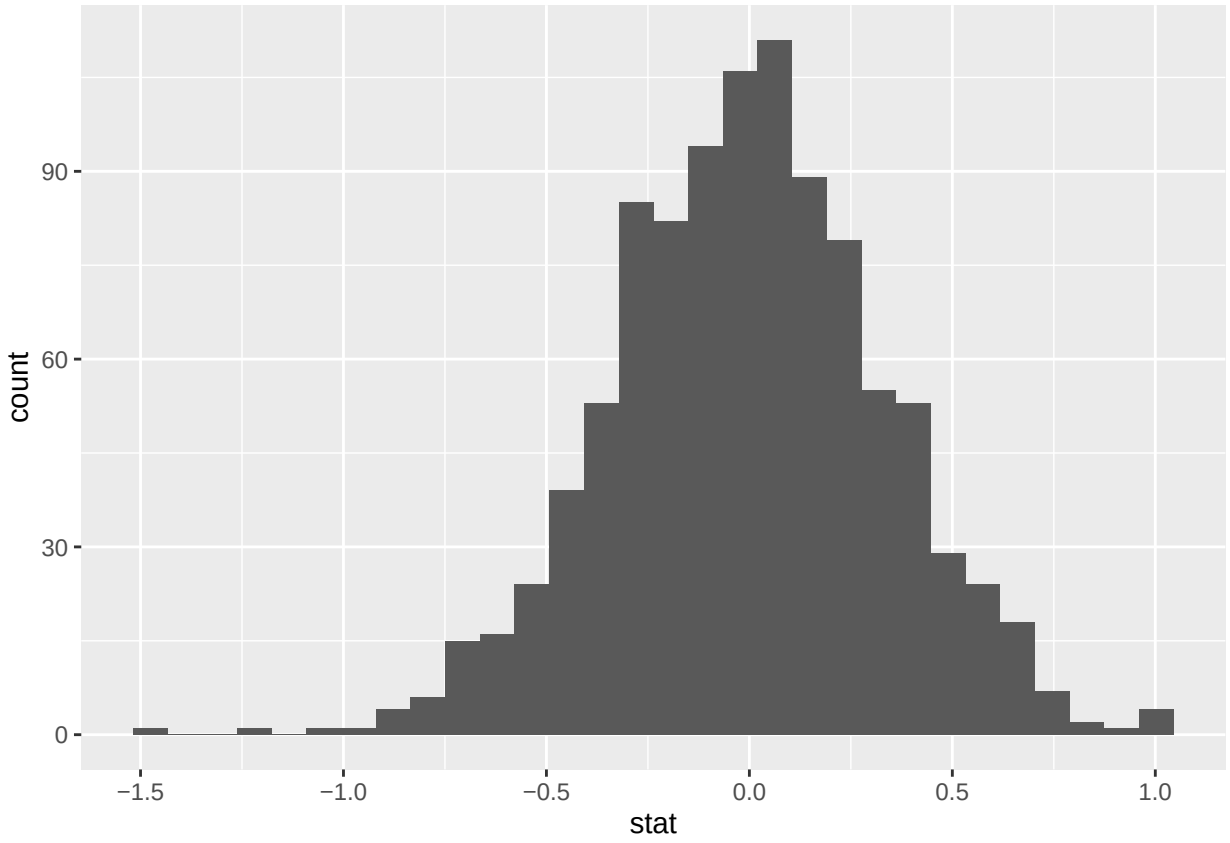
```
null_dist <- yrbss %>%
  drop_na(physical_3plus) %>%
  specify(weight ~ physical_3plus) %>%
  hypothesize(null = "independence") %>%
  generate(reps = 1000, type = "permute") %>%
  calculate(stat = "diff in means", order = c("yes", "no"))
```

Here, `hypothesize` is used to set the null hypothesis as a test for independence. In one sample cases, the `null` argument can be set to "point" to test a hypothesis relative to a point estimate.

Also, note that the `type` argument within `generate` is set to `permute`, whichis the argument when generating a null distribution for a hypothesis test.

We can visualize this null distribution with the following code:

```
ggplot(data = null_dist, aes(x = stat)) +
  geom_histogram()
```



6. How many of these `null` permutations have a difference of at least `obs_stat`?

Insert your answer here

```
null_dist %>%
  filter(stat >= obs_diff$stat) %>%
  nrow()
```

```
## [1] 0
```

0 of them.

Now that the test is initialized and the null distribution formed, you can calculate the p-value for your hypothesis test using the function `get_p_value`.

```
null_dist %>%
  get_p_value(obs_stat = obs_diff, direction = "two_sided")
```

```
## # A tibble: 1 x 1
##   p_value
```

```
##     <dbl>
## 1       0
```

This the standard workflow for performing hypothesis tests.

7. Construct and record a confidence interval for the difference between the weights of those who exercise at least three times a week and those who don't, and interpret this interval in context of the data.

```
yrbss %>%
  drop_na(physical_3plus, weight) %>%
  specify(weight ~ physical_3plus) %>%
  generate(reps = 1000, type = "bootstrap") %>%
  calculate(stat = "diff in means", order = c("yes", "no")) %>%
  get_ci(level = 0.95, type = "percentile")
```

```
## # A tibble: 1 x 2
##   lower_ci upper_ci
##      <dbl>    <dbl>
## 1     1.14     2.43
```

The 95% confidence interval for $\mu_{yes} - \mu_{no}$ is approximately (1.2 kg, 2.5 kg). We can be 95% confident that the average weight of high school students who exercise regularly at least three days a week is 1.2 to 2.5 kg greater than the average weight of high school students who do not exercise regularly. This interval doesn't include 0, and that supports the result of hypothesis testing, where we rejected H_0.

More Practice

8. Calculate a 95% confidence interval for the average height in meters (`height`) and interpret it in context.

Insert your answer here

```
height_ci <- yrbss %>%
  drop_na(height) %>%
  specify(response = height) %>%
  generate(reps = 1000, type = "bootstrap") %>%
  calculate(stat = "mean") %>%
  get_ci(level = 0.95, type = "percentile")

sprintf("%.5f, %.5f", height_ci$lower_ci, height_ci$upper_ci)
```

```
## [1] "1.68940, 1.69312"
```

The 95% confidence interval for mean height is approximately 1.689 m, 1.693 m. We are 95% confident that the true average height of U.S. high school students is between about 1.689 and 1.693 meters.

9. Calculate a new confidence interval for the same parameter at the 90% confidence level. Comment on the width of this interval versus the one obtained in the previous exercise.

Insert your answer here

```
height_ci90 <- yrbss %>%
  drop_na(height) %>%
  specify(response = height) %>%
  generate(reps = 1000, type = "bootstrap") %>%
  calculate(stat = "mean") %>%
  get_ci(level = 0.90, type = "percentile")

sprintf("%.5f", height_ci90$lower_ci)
```

```
## [1] "1.68962"
```

```
sprintf("%.5f", height_ci90$upper_ci)
```

```
## [1] "1.69273"
```

The 90% confidence interval is approximately 1.69 m, 1.693 m. It is slightly narrower than the 95% interval. A lower confidence level uses a smaller critical value, so we get a narrower interval and we gain precision but are less confident that it captures the true mean.

10. Conduct a hypothesis test evaluating whether the average height is different for those who exercise at least three times a week and those who don't.

Insert your answer here

Hypotheses: $H_0 : \mu_{height,yes} - \mu_{height,no} = 0$ and $H_A : \mu_{height,yes} - \mu_{height,no} \neq 0$, tested at $\alpha = 0.05$.

```
obs_diff_h <- yrbss %>%
  drop_na(physical_3plus, height) %>%
  specify(height ~ physical_3plus) %>%
  calculate(stat = "diff in means", order = c("yes", "no"))

null_dist_h <- yrbss %>%
  drop_na(physical_3plus, height) %>%
  specify(height ~ physical_3plus) %>%
  hypothesize(null = "independence") %>%
  generate(reps = 1000, type = "permute") %>%
  calculate(stat = "diff in means", order = c("yes", "no"))

null_dist_h %>%
  get_p_value(obs_stat = obs_diff_h, direction = "two_sided")
```

```
## # A tibble: 1 x 1
##   p_value
##     <dbl>
## 1       0
```

We reject H_0: there is strong evidence that the average height differs between the two groups, with the more active students being taller on average. The most likely reason is that boys are taller and more likely to be physically active, so this is an association and not evidence that exercise changes height.

11. Now, a non-inference task: Determine the number of different options there are in the dataset for the `hours_tv_per_school_day` there are.

Insert your answer here

```
yrbss %>%
  count(hours_tv_per_school_day)
```

```
## # A tibble: 8 x 2
##   hours_tv_per_school_day     n
##   <chr>                   <int>
## 1 1                        1750
## 2 2                        2705
## 3 3                        2139
## 4 4                        1048
## 5 5+                       1595
## 6 <1                       2168
## 7 do not watch             1840
## 8 <NA>                      338
```

```
n_distinct(yrbss$hours_tv_per_school_day, na.rm = TRUE)
```

```
## [1] 7
```

There are 7.

12. Come up with a research question evaluating the relationship between height or weight and sleep. Formulate the question in a way that it can be answered using a hypothesis test and/or a confidence interval. Report the statistical results, and also provide an explanation in plain language. Be sure to check all assumptions, state your α level, and conclude in context.

Insert your answer here

Research question: Do high schoolers who get short sleep on school nights, fewer than 7 hours, have a different average weight than those who get 7 or more hours?

Let μ_{short} and μ_{enough} be the mean weights of those sleeping under 7 hours and 7+ hours.

- $H_0 : \mu_{short} - \mu_{enough} = 0$
- $H_A : \mu_{short} - \mu_{enough} \neq 0$
- $\alpha = 0.05$

```
set.seed(600)
yrbss_sleep <- yrbss %>%
  mutate(sleep_group = ifelse(school_night_hours_sleep %in% c("<5", "5", "6"),
                              "short", "normal")) %>%
  drop_na(sleep_group, weight)

yrbss_sleep
```

```
## # A tibble: 12,579 x 15
##      age gender grade hispanic race                     height weight helmet_12m
##    <int> <chr>  <chr> <chr>    <chr>                     <dbl>  <dbl> <chr>
##  1    15 female 9     hispanic Native Hawaiian or Othe~   1.73   84.4 never
##  2    15 female 9     not      Black or African Americ~   1.6    55.8 never
##  3    15 female 9     not      Black or African Americ~   1.5    46.7 did not r~
##  4    15 female 9     not      Black or African Americ~   1.57   67.1 did not r~
##  5    15 female 9     not      Black or African Americ~   1.65  132.  did not r~
##  6    14 male   9     not      Black or African Americ~   1.88   71.2 never
##  7    15 male   9     not      Black or African Americ~   1.75   63.5 never
##  8    15 male   10    not      Black or African Americ~   1.37   97.1 did not r~
##  9    15 female 9     not      Black or African Americ~   1.68   69.8 did not r~
## 10    15 female 9     not      Black or African Americ~   1.65   66.7 did not r~
## # i 12,569 more rows
## # i 7 more variables: text_while_driving_30d <chr>, physically_active_7d <int>,
## #   hours_tv_per_school_day <chr>, strength_training_7d <int>,
## #   school_night_hours_sleep <chr>, physical_3plus <chr>, sleep_group <chr>
```

```
set.seed(600)
yrbss_sleep %>%
  group_by(sleep_group) %>%
  summarise(mean_weight = mean(weight), sd_weight = sd(weight), n = n())
```

```
## # A tibble: 2 x 4
##   sleep_group mean_weight sd_weight     n
##   <chr>             <dbl>     <dbl> <int>
## 1 normal             67.4      16.4  7846
## 2 short              68.7      17.7  4733
```

```r
set.seed(600)
obs_diff_s <- yrbss_sleep %>%
  specify(weight ~ sleep_group) %>%
  calculate(stat = "diff in means", order = c("short", "normal"))
```

```r
set.seed(600)
null_dist_s <- yrbss_sleep %>%
  specify(weight ~ sleep_group) %>%
  hypothesize(null = "independence") %>%
  generate(reps = 1000, type = "permute") %>%
  calculate(stat = "diff in means", order = c("short", "normal"))
```

```r
set.seed(600)
null_dist_s %>% get_p_value(obs_stat = obs_diff_s, direction = "two_sided")
```

```
## # A tibble: 1 x 1
##   p_value
##     <dbl>
## 1       0
```

```r
set.seed(600)
yrbss_sleep %>%
  specify(weight ~ sleep_group) %>%
  generate(reps = 1000, type = "bootstrap") %>%
  calculate(stat = "diff in means", order = c("short", "normal")) %>%
  get_ci(level = 0.95, type = "percentile")
```

```
## # A tibble: 1 x 2
##   lower_ci upper_ci
##      <dbl>    <dbl>
## 1    0.656     1.93
```

The sampling is random, the sample size is less than 10% of the population, the two samples are of different people, and both samples are fairly large, approximately 4,733 for short sleepers and 6,748 for those with at least 7 hours. Weight is right-skewed, but with samples this large, the CLT applies to the distribution of the difference.

The mean weight of short sleepers was 68.7 kg, while that of the people who slept for at least 7 hours was 67.3 kg, with an observed difference of 1.40 kg. The value of the p-value based on the permutation test is close to zero, and hence is less than $\alpha = 0.05$. The 95% CI for the difference is (0.656 kg, 1.93 kg).

Statistical evidence proves that the students sleeping for less than 7 hours have greater mean weights than those sleeping for 7 or more hours, which differs from the former ones in approximately 1-2 kg. This deviation is relatively small as compared to the variance of weight, and it should be noted that this is only the observational survey and not the experiment, and hence we cannot conclude causality because other variables such as age and sex could be influencing this result.
